Unveiling Sentiment Weighted Network Analysis (SWENA) with ERGM in Ottoman-Turkish Memoirs: A Hybrid Causal Analysis

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Abstract

This study proposes a hybrid analytical framework that links automated sentiment extraction to inferential network modelling in Historical Network Research (HNR). While the field has robust methods for reconstructing the existence of ties between historical actors (Ahnert et al. 2021), it has fewer for analyzing their affective valence-whether a recorded relationship carried trust, indifference, or hostility. The framework addresses this gap by combining a deep-learning sentiment pipeline, Cross-individual Sentiment Analysis (CISA; İlter et al. 2026), with Exponential Random Graph Models (ERGM), and illustrates the workflow on a corpus of memoirs from the late Ottoman and early Republican periods.
CISA classifies a memoirist's expressed attitude toward each named individual as negative, neutral, or positive. The resulting outcome therefore represents authorial sentiment, conditioned by the temporal distance and the self-legitimizing conventions of the memoir form, rather than a directly observed social relationship. The SWENA module (Sentiment-Weighted Network Analysis) then assembles these signed attitudes into an actor network. Through textual co-occurrence weighting and a polarity-ratio filter, it attenuates the predominantly neutral narrative register that typically obscures the affective signal in textual archives. The extraction procedure draws on structural balance theory (Heider 1946; Cartwright and Harary 1956) and signed blockmodelling (Doreian and Mrvar 1996, 2009) to identify coherent factions.

The second phase situates the signed network as a statistical dependent variable rather than a descriptive output. Within an ERGM framework (Robins et al. 2007; Lusher et al. 2013), tie polarity is modelled as a function of two ordinarily separate classes of network explanation. The first is exogenous: nodal attributes such as shared education, military rank, or factional membership, which capture socio-political homophily (McPherson et al. 2001). The second is endogenous: the structural dependencies intrinsic to signed networks, including the triadic balance configurations predicted by Heider's theory. The joint modelling of these classes builds on recent advances in signed-network inference (Yap and Harrigan 2015; Fritz et al. 2025), which the present framework adapts.

Where existing signed ERGMs model observed cooperation or conflict, here the dependent structure is the retrospectively expressed attitude of an author. The framework can thus ask how exogenous attributes condition the way affective ties close into balanced configurations. For instance, it can test whether shared Committee of Union and Progress (CUP) membership raises the raises the likelihood that the triad among Enver, Talat, and Cemal Pashas

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closes into a positively balanced configuration, relative to triads lacking a common factional origin. The analytical question thus shifts from the descriptive structure of the network to the joint attribute-and-structure mechanisms associated with its sign configuration.

By integrating automatically extracted sentiment with probabilistic network models, the framework offers HNR a replicable procedure for moving from the description of affective relations toward their statistical analysis, while remaining explicit about the interpretive limits of narrative sources.

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